MILITARY SPECIFICATION

DIVING EQUIPMENT SETS

PACKAGING OF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for preservation, packaging, packing, and marking of Diving Equipment Sets, Open Circuit Scuba (SC 4220-97-CL-E09), Closed Circuit Scuba (SC 4220-97-CL-E08), Individual Swimmer Support, Scuba (SC 4220-97-CL-E11), and Scuba Diving Support (SC 4220-97-CL-E10) and sized components.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

UU-T-81	- Tags, Shipping and Stock.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-T-60	- Tape: Packaging, Waterproof.

Military

MIL-C-104	- Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	- Preservation-Packaging, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-E-10062	- Engines: Preparation for Shipment and Storage of.

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STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-281	- Automobiles, Trucks, Truck-Tractors, Trailers and Trailer Dollies; Preservation and Packaging of.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, DC 20036.)

3. REQUIREMENTS

3.1 Preproduction pack. The supplier shall furnish a preproduction pack for examination within the time frame specified (see 6.2) to prove prior to starting production packaging and packing, that the applied preservation,

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packaging, packing, and marking comply with the requirements of this specification. Examination shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.2 Materials. Materials shall be as specified herein and in applicable specifications and standards. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification. Materials shall be free of defects.

3.3 Preservation and packaging. Preservation and packaging shall be Level A or C as specified (see 6.2).

3.3.1 Level A.

3.3.1.1 Methods of preservation. Methods of preservation indicated herein by symbols, shall be in accordance with the applicable symbols listed in MIL-P-116.

3.3.1.2 Diving equipment sets.

3.3.1.2.1 Open circuit scuba.

3.3.1.2.1.1 Tank block. Each tank block shall be preserved Method III, in a close-fitting box conforming to PPP-B-636, Class Domestic, Variety SW, Grade 175, style optional.

3.3.1.2.1.2 Air pressure regulator, tank pressure gage with fittings and test pressure gage. Each air pressure regulator, each tank pressure gage with fittings, and each test pressure gage shall be preserved Method III, using the vendor's carton or a close-fitting box conforming to PPP-B-566, Variety 1, style, type, and class optional.

3.3.1.2.2 Closed circuit scuba.

3.3.1.2.2.1 Backpack rebreather assembly and test pressure gages. The backpack rebreather assembly and test pressure gages shall be placed in the case. Components shall be positioned to provide a cushioned surface for glass surfaces of display gages and test pressure gages and shall be immobilized within the case. The case lid shall be closed and secured.

3.3.1.2.3 Individual swimmer support, scuba.

3.3.1.2.3.1 Basic components. Each component of the set shall be preserved Method III, using the vendor's carton or a close-fitting box conforming to PPP-B-566, Variety 1, style, type, and class optional or PPP-B-636, Class Domestic, Variety SW, Grade 125, style optional.

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3.3.1.2.3.2 Sized components. Each sized component shall be preserved Method IC-2, using the vendor's carton or a close-fitting box conforming to PPP-B-566, Variety 1, class, type, and style optional or Method III, using a close-fitting box conforming to PPP-B-636, W6c, style optional. Box closure shall be in accordance with the appendix to the box specification, Method V.

Note: Basic components and sized components fabricated of rubber shall be dusted with talc before preservation of the component is accomplished.

3.3.1.2.4 Scuba diving support.

3.3.1.2.4.1 Gas purity detector and protective weapon kit. Each gas purity detector and each protective weapon kit shall be preserved Method III, using the vendor's carton or a close-fitting box conforming to PPP-B-566, Variety 1, style, type, and class optional.

3.3.1.2.4.2 Underwater photographic kit. Each component of the underwater photographic kit shall be preserved Method III, using the vendor's carton for the accessories and the vendor's polystyrene container for the camera or a close-fitting box conforming to PPP-B-566, Variety 1, style, type, and class optional. The box containing the camera shall be fitted with molded polystyrene inserts or resilient foam cushioning to control movement of the camera within the container.

3.3.1.2.4.3 Decompression chamber. The decompression chamber shall not require preservation or packaging.

3.3.1.2.4.4 Compressors. Each 15-20 cfm compressor and each 3-5 cfm compressor shall be preserved Method IIa by bolting directly through the barrier material to the crate base specified in 3.4.1.1.2.2. Before inclosing the compressors in the Method II barrier material, the engine and engine accessories shall be preserved and packaged in accordance with MIL-E-10062, Level A, Preparation Type II, Method II. Compressor drivebelts shall be released from tension. A depreservation guide shall be prepared for each compressor. The depreservation guide shall be preserved Method IC-1, and shall be attached to the compressor. Unless otherwise specified (see 6.2), "DA Form 2258 - Depreservation Guide For Vehicles and Equipment" shall be used (see 6.4).

Note: Contact preservatives or preservative oils shall not be applied to any internal surfaces of the compressor unit during preservation of the engine. Only approved compressor lubricants shall be used for operation of the compressor unit.

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3.3.1.2.4.5 Cascade manifold system. All external openings into the cascade manifold recharge system shall be sealed with barrier material conforming to MIL-B-121, Type I, Grade A, Class 1. The barrier material shall be secured in place with tape conforming to PPP-T-60, Type IV. The protective metal cover shall be positioned to cover the control panel and secured. The unit shall be tagged to indicate that all openings into the unit are sealed and that the seals must be removed prior to putting the manifold system into service. Tags shall conform to UU-T-81, Type A. The tag and printing thereon shall be waterproofed in accordance with MIL-STD-129. The trailer chassis components shall be preserved and packaged in accordance with MIL-STD-281, Level A for mobile shipment.

3.3.1.2.4.6 Sized components. Each sized dress set shall be preserved Method IC-2 using the vendor's carton or a close-fitting box conforming to PPP-B-566, Variety 1, class, type, and style optional or Method III using a close-fitting box conforming to PPP-B-636, W6c, style optional. Box closure shall be in accordance with the appendix to the box specification, Method V.

Note: Components of the dress set fabricated of rubber shall be dusted with talc before preservation of the dress set is accomplished.

3.3.1.3 Technical publications. Technical publications furnished for components of each applicable diving set shall be preserved Method IC-1.

3.3.1.4 Consolidated packaging of sets. The packaged components and applicable publications of each open circuit set, the case containing the components, and applicable publications of each closed circuit set, the packaged components and applicable publications of each individual swimmer support set and components of each scuba diving support set that are preserved Method III and applicable set publications shall be consolidated packaged in a close-fitting box conforming to PPP-B-636, V3c, W5c or W6c, as applicable, style optional. Cushioning and blocking shall be provided to control movement of components within the container. Box closure shall be in accordance with the appendix to the box specification, Method V.

Note: Each consolidated packaged box shall contain only those components and publications that are applicable to a specific set.

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3.3.1.5 Intermediate packaging of sized components. Packaged sized components of like description shall be intermediate packaged together, not to exceed the weight limitation, in close-fitting boxes conforming to PPP-B-636, Class Weather Resistant, style optional. Cushioning and blocking shall be provided to control movement of components within the container. Box closure shall be in accordance with the appendix to the box specification, Method V.

3.3.2 Level C. Unless otherwise specified herein, the components and applicable technical publications of each open circuit set, each closed circuit set, each individual swimmer support set, each scuba diving support set or sized components shall be preserved and packaged to afford protection against deterioration and damage from the supplier to the initial destination. The supplier's standard commercial packaging is acceptable when it fulfills these requirements. When specified (see 6.2), the following alternate preservation shall be performed.

3.3.2.1 Alternate preservation. Each compressor engine fuel system, combustion chambers and valves and coolant system when applicable, shall be preserved in accordance with MIL-E-10062, Level A, Type I or II as applicable, Method I (see 6.2).

3.4 Packing. Packing shall be Level A, B or C as specified (see 6.2).

3.4.1 Level A.

3.4.1.1 Diving equipment sets.

3.4.1.1.1 Open circuit scuba, closed circuit scuba and individual swimmer support scuba. Each complete open circuit scuba diving set, each complete closed circuit scuba diving set and each complete individual swimmer support scuba diving set, packaged as specified in 3.3 shall be packed in a close-fitting box conforming to PPP-B-601, Overseas Type, style optional. Strapping shall be zinc coated.

Note: Each packed box shall contain only packaged components applicable to a specific set.

3.4.1.1.2 Scuba diving support.

3.4.1.1.2.1 Boxed components. Components of the scuba diving support set preserved Method III and consolidated packaged as specified in 3.3.1.4 shall be packed in a close-fitting box conforming to PPP-B-601, Domestic Type, style optional. Strapping shall not be required.

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3.4.1.1.2.2 Crated components. Components comprising a complete scuba diving support set, including the components boxed in accordance with 3.4.1.1.2.1 but excluding the trailer mounted cascade manifold system, shall be packed in a crate conforming to MIL-C-104, Type I, Class 1 or 2, Style a. The components shall be arranged to provide a crate of minimum cube. Inspection doors shall be provided and shall be placed to provide for inspection of the humidity indicators of the items preserved Method II. Clearance shall be provided between the Method II barriers and other anchored components to prevent abrasion of the barrier material. Blocking, bracing, anchoring, closure, and strapping shall be in accordance with the crate specification and MIL-STD-1186. Strapping shall be zinc coated.

3.4.1.1.2.3 Cascade manifold system. The trailer mounted cascade manifold system preserved and packaged in accordance with 3.3 for mobile shipment shall not require packing.

Note: The cascade manifold system shall be shipped separate but at the same time as the crated components.

3.4.1.2 Sized components. Sized components of like description preserved and packaged in accordance with 3.3 shall be packed, not to exceed the weight limitation, in close-fitting boxes conforming to PPP-B-601, Overseas Type, style optional. Strapping shall be zinc coated.

3.4.2 Level B.

3.4.2.1 Diving equipment sets.

3.4.2.1.1 Open circuit scuba, closed circuit scuba and individual swimmer support scuba. Packing shall be as specified for Level A, except that boxes shall be Domestic Type and strapping will not be required to be zinc coated.

3.4.2.1.2 Scuba diving support. Level B packing is not applicable to this set.

3.4.2.2 Sized components. Packing shall be as specified for Level A, except that boxes shall be Domestic Type and strapping will not be required to be zinc coated.

3.4.3 Level C. Each complete open circuit scuba diving set, each complete closed circuit scuba diving set, each complete individual swimmer support scuba diving set, each complete scuba diving support set or sized components,

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packaged as specified in 3.3, shall be packed or prepared for shipment in a manner to assure carrier acceptance and safe delivery to destination at the lowest ratings in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules. Wheels shall be free to rotate on trailers prepared for mobile shipment.

3.5 Marking. Marking shall be in accordance with MIL-STD-129.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material inspection. The supplier is responsible for insuring that supplies and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. Inspections shall be classified as follows:

(a) Preproduction pack inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction pack inspection.

4.3.1 Examination. The preproduction pack shall be examined for the defects listed in 4.4.4. Presence of one or more defects shall be cause for rejection of the preproduction pack.

4.4 Quality conformance inspection.

4.4.1 Inspection stages. Inspection shall be in three stages as follows:

(a) The first stage shall include inspection of procedures, methods, materials, containers, and packaging marking prior to placing the packages into consolidated and intermediate containers.

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(b) The second stage shall include inspection of closure and marking of consolidated and intermediate containers and the arrangement, cushioning, blocking, bracing, and anchoring of packaged components within the shipping containers.
(c) The third stage shall include inspection of the closed, strapped, and marked shipping containers and trailers prepared for shipment.

4.4.2 Unit of product. For the purpose of inspection, all packages, pack, and components, prepared for any applicable inspection stage that will comprise a complete pack for an applicable set, shall be considered a unit of product.

4.4.3 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.4 Examination. Samples selected in accordance with 4.4.3 shall be examined for the defects marked "X" for the applicable level in Table I. Acceptable Quality Level (AQL) shall be 2.5 percent defective.

Table I. Classification of Defects.

No.	Defect	Level A	Level B	Level C
101.	Materials, methods, or containers not as specified. Each incorrect material method or container shall constitute one defect.	X	X	-
102.	Components of the closed circuit scuba set not positioned to provide cushioning for glass surfaces of gages and the components not immobilized.	X	-	-
103.	Fiberboard boxes not closed in accordance with Method V closure as specified.	X	-	-
104.	Rubber components not dusted with talc as specified.	X	-	-
105.	Box inserts or cushioning not provided as specified.	X	-	-
106.	Compressor engines and engine accessories not preserved as specified.	X	-	X
107.	Compressor drivebelts not released from tension as specified.	X	-	-
108.	Depreservation guide not prepared and attached to the compressor as specified.	X	-	-
109.	Openings into the cascade recharge manifold system not sealed and the unit tagged as specified.	X	-	-
110.	Protective metal cover not positioned and secured on the cascade recharge manifold system as specified.	X	-	-

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Table I. Classification of Defects (Cont'd)

No.	Defect	Level A	Level B	Level C
111.	Trailer chassis components of the cascade recharge manifold system not preserved and packaged as specified.	X	-	-
112.	Consolidated packaging of set and components not as specified.	X	-	-
113.	Intermediate packaging of sized components not as specified.	X	-	-
114.	Strapping not zinc coated as specified.	X	-	-
115.	Component not arranged in the crate to produce a crate of minimum cube.	X	-	-
116.	Inspection doors not provided in the crate as specified.	X	-	-
117.	Clearance not provided for barriers of items preserved Method IIa and secured in the crate.	X	-	-
118.	Blocking, bracing, and anchoring within the crate not as specified.	X	-	-
119.	Packing of sets not as specified.	X	X	-
120.	Marking illegible, incomplete, incorrect, or missing.	X	X	X

5. PREPARATION FOR DELIVERY

(This section not applicable to this specification).

6. NOTES

6.1 Intended use. This specification is intended to be used for direct reference in contracts or purchase orders for the preservation, packaging, packing, and marking of Diving Equipment Sets, Open Circuit Scuba, Closed Circuit Scuba, Individual Swimmer Support Scuba, Scuba Diving Support, and sized components.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of the preproduction pack (see 3.1).
(c) Level of preservation and packaging and level of packing required (see 3.3 and 3.4).

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(d) Kits of diving equipment set to be packaged and packed (see 3.3.1.2.1, 3.3.1.2.2, 3.3.1.2.3, and 3.3.1.2.4).
(e) When engine components are to be preserved for Level C (see 3.3.2).
(f) When depreservation guide form other than as specified is required (see 3.3.1.2.4.4).

6.3 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the supplier of his obligation to preserve, package, pack, and mark the diving equipment set components in accordance with this specification.

6.4 Depreservation guide. The contracting officer should arrange to furnish the necessary copies of DA Form 2258 when requested by the supplier (see 3.3.1.2.4.4).

Custodian:

Army - ME

Preparing activity:

Army - ME

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE

MIL-D-52838(ME) Diving Equipment Gen, Packaging Of

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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